

cocotb scoreboard

A scoreboard checks that what the DUT actually did matches what you expected, transaction by transaction

Expect queue and observe

- Order matters, the queue is a FIFO
- Push expect hex A five before the DUT responds, then observe A five when data is valid
- One match empties the queue
- Push two expects in order and observe both in order for two passes
- Observe the wrong byte and you fail without popping
- Observe when the queue is empty and you fail too, there was nothing left to check against

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

cocotb scoreboard sketch

Expect queue vs observed actuals — match pops **PASS**, mismatch **FAIL**. Starter: expect
0xA5, observe 0xA5 → PASS.

Starter example: expect 0xA5, observe 0xA5 → **PASS**, queue empty.

Load starter example

Challenges 0 / 22 cleared

Quiz: **expect**: scoreboard.expect(value)...

☐ pushes an expected value onto the compare queue

☐ immediately fails the test

☐ starts the clock generator

☐ clears the VCD dump

Show hint

Check

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Core ideas

expect
Push golden values onto FIFO queue.

observe
Compare actual when transaction done.

match
PASS and pop front on match.

mismatch
FAIL — queue kept for debug.

Browser lab starter

learn_cocotb — cocotb scoreboard

Real cocotb practice

- In the real cocotb track
- Draw a three-step timeline: push expect, DUT responds, compare and pop or fail
- Write pseudocode for expect push and compare on one byte
- Optional: open the lab's source sketch and read how match and mismatch are commented
- This module is expect-versus-observe literacy, not a full pyuvvm scoreboard yet

Pitfalls to watch

- Do not compare before the transaction is complete
- Do not pop the queue on failure; you need the expected value left for debug
- A common miss is pushing expects out of order when the DUT returns in order
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and predict pass before you observe
- On paper, sketch push, observe, and pop for two expects in sequence
- When you are ready